Queues

DSA27 — Lecture 07 · Data Structures and Algorithms

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> ***How to read this document.*** *This is the handout for Lecture 07. It holds everything on the slides, plus what I said out loud. A queue looks like a stack with one word changed, and that one word,* first *instead of* last*, is enough to break the obvious design. This week you measure that break yourself, then fix it with the ring buffer, the most useful trick in the course that fits in one line of arithmetic.*
>
> *Slides:* `DSA27-L07-slides.pdf` *· Code:* `dsa/queue.py` *· Tests:* `tests/test_stack_queue.py` *(the queue half) · Interactive demo: Queue Lab · Practise online: Week 7 quiz and written practice (in Arabic) · timed Lecture 7 test · midterm simulator*

Where We Are

One end was easy. Now we need both.

Week 6: the **stack** touches **one** end. Put the top at the cheap end, done.

Today: the **queue** adds at one end and removes at the **other**.

- A dynamic array is cheap at the **end** only.
- A singly linked list is cheap at the **head** (and, with a tail, for adding at the end).

So which end goes where, and what happens to the expensive end?

Lecture 06 closed with a promise: this is the week your `DynamicArray` finally meets an operation it is bad at. A stack could hide the array's weak end simply by never using it. A queue cannot: whatever you do, one of its two operations lands on the front of the array.

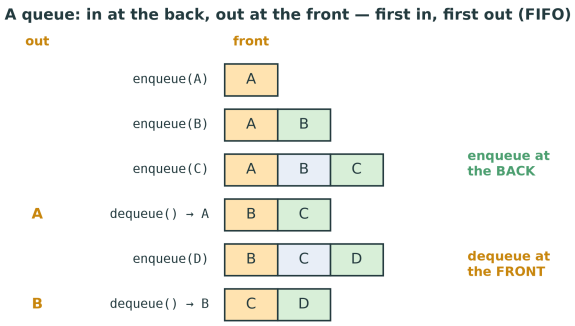
Queue in Arabic: الطابو�.

Today

1. The Queue ADT: enqueue, dequeue — FIFO
2. The obvious design, `SlowQueue`, and its $O(n)$ dequeue
3. The ring buffer: `CircularQueue`, every operation $O(1)$
4. Full or empty? — and growing a ring without scrambling it
5. The linked queue, and a queue made of two stacks
6. Measured: the gap is the lesson
7. Where queues are used — and the deque

The Queue ADT

First in, first out



Operation	Meaning	Cost
`enqueue(x)`	add x at the **back**	$O(1)$
`dequeue()`	remove and return the **front** (`IndexError` if empty)	$O(1)$
`is_empty()`, `len(q)`		$O(1)$

That table is the ADT: operations, meaning, cost, and nothing about storage — exactly as with the stack. Many textbooks add a peek (also called `front` or `first`) that returns the front without removing it; the exercise does not need it, but it is one line once the rest works.

Everyday queues: a line at a bank or a bakery, where the first to arrive is the first served; a printer's job list; the messages waiting for a WhatsApp server to deliver them; the keys you press while a program is busy, which arrive in the order you pressed them. The rule is **fairness in order of arrival** — and that is exactly what a stack does not give you (W6-M20).

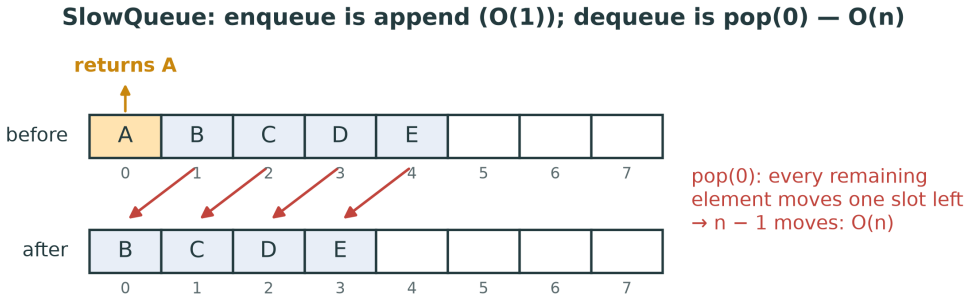
Empty queues. Dequeuing an empty queue raises `IndexError`, like popping an empty stack. `tests/test_stack_queue.py::test_dequeue_on_empty_raises` checks it for both classes.

Stack against queue, in one line each:

	Adds at	Removes at	Order out
Stack	the top	the **same** end	reversed (LIFO)
Queue	the back	the **other** end	preserved (FIFO)

The Obvious Design

`SlowQueue`: a `DynamicArray`, as it is



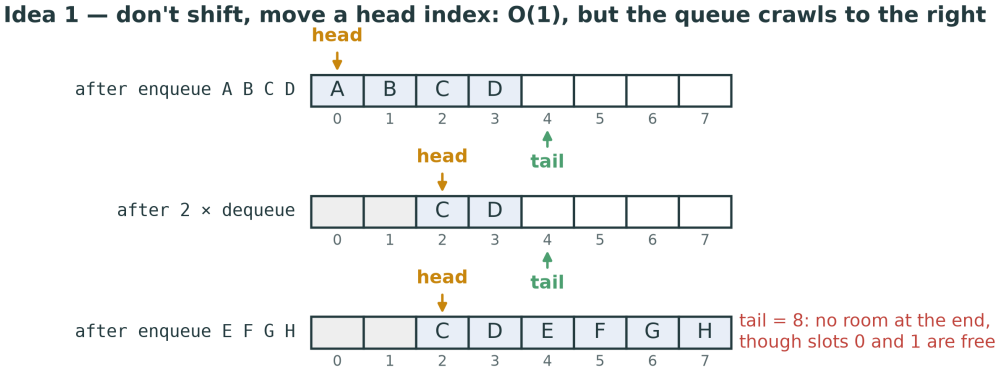
- enqueue = append at the end: amortised $O(1)$.
- dequeue = `pop(0)`: every remaining element shifts left — $O(n)$.

This is `SlowQueue` in `dsa/queue.py`, and it is a correct queue: it passes every test. The class is deliberately named for its flaw, because you are going to measure it.

Could we swap the ends? Put the back at index 0 and the front at the end: then dequeue is `pop()` at the end, $O(1)$, but enqueue becomes `insert_at(0, x)`, which shifts everything — $O(n)$ again. With a plain array, **whichever** end is the front, one of the two operations costs $O(n)$. That is the difference from the stack, which could simply put all its work at the cheap end.

Python's own `list` has the same problem: `list.pop(0)` is $O(n)$. That is why Lab 03 told you to use `collections.deque` for a queue, and promised you would measure the difference this week.

Idea 1: don't shift — move the front



Keep a `head` index. Dequeue reads `block[head]` and adds 1 to `head`: $O(1)$.

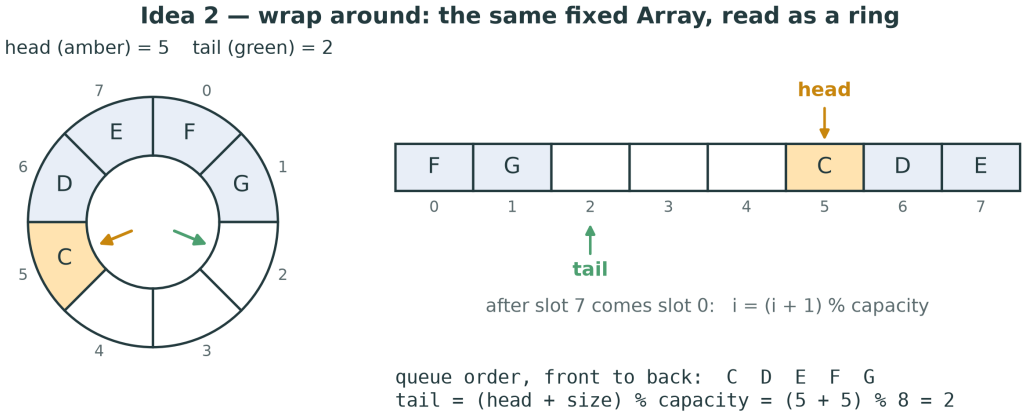
But the used part crawls right, and the slots it leaves behind are never used again.

Dequeue is now $O(1)$ — nothing moves. The price is that dequeued slots at the front are dead. In the figure, eight enqueues and two dequeues leave only six items in an eight-slot block, yet there is no room for a ninth at the end. A long-running queue with this design grows its block for ever, even if it never holds more than three items at a time — a memory leak by design.

You could compact now and then — shift everything back to index 0 when the dead space gets large — and that is a legitimate design with an amortised argument like Lecture 04's. But there is a neater answer that never moves anything.

The Ring Buffer

Idea 2: wrap around



After the last slot comes slot 0 again:

- **front** at head; **back** at `(head + size) % capacity`
- every step forward is `(i + 1) % capacity`

The block is still a plain, fixed `Array`. What changes is how we **read** it: index arithmetic *modulo the capacity* makes the end of the block join up with the beginning, so the slots freed at the front are reused as soon as the back reaches them. This is called a **ring buffer** or **circular buffer**, and it is everywhere in systems programming: keyboard buffers, network cards, audio streaming, logging the last N events. Wherever data arrives at one rate and leaves at another, there is probably a ring buffer between them.

`dsa/queue.py` stores three things besides the block: `_capacity`, `_head` (the index of the front item) and `_size` (how many items are in the queue). It **does not** store the tail; the tail is computed:

$$\text{tail} = (\text{head} + \text{size}) \bmod \text{capacity}$$

In the figure, head = 5 and size = 5 in a block of 8, so the tail — the slot the next enqueue writes to — is (5 + 5) mod 8 = 2. The queue, front to back, is C D E F G: slots 5, 6, 7, then round to 0 and 1. `__iter__` in `dsa/queue.py` is given to you and walks exactly this way: `(head + offset) % capacity` for each offset from 0 to size − 1.

The two operations

enqueue(x)

1. full? — see the next slides

2. write x at `(head + size) % capacity`
3. `size += 1`

dequeue()

1. empty? — `IndexError`
2. read the value at `head`; clear that slot
3. `head = (head + 1) % capacity`; `size -= 1`

Nothing moves. Every step is a little arithmetic: $O(1)$.

Those six steps are the whole exercise, apart from the challenge; turning them into Python is your job. Two details are worth saying out loud.

Why clear the slot? Setting the dequeued slot back to `None` is not needed for the queue to be correct: that slot is outside the live part of the ring and will be overwritten later. But until then the block would still hold a reference to the dequeued object, and Python's garbage collector could not free it. Clearing it is the same care as `DynamicArray.pop` setting the vacated slot to `None`.

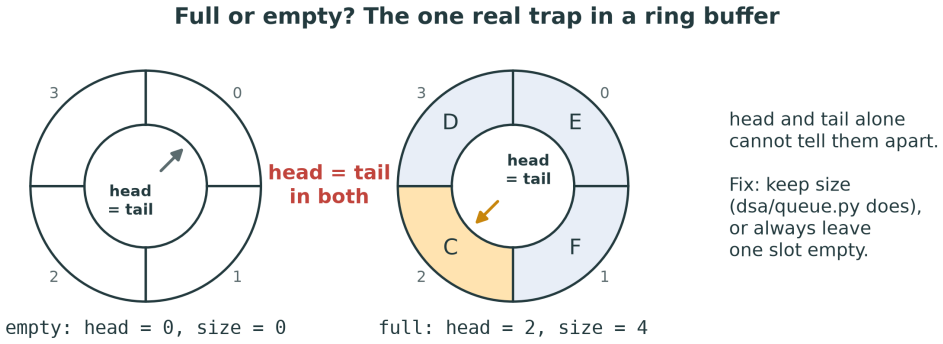
The modulo is the whole trick. Without it, `head + 1` would walk off the end of the `Array` and raise `IndexError`. `test_circular_queue_wraps_around` fills a ring of capacity 3, dequeues once and enqueues again: the new value must land in the slot at index 0, which only the `%` makes possible.

Trace: capacity 4

Operation	Block	head	size	Returns
start	_ _ _ _	0	0	
enqueue A, B, C	A B C _	0	3	
dequeue	_ B C _	1	2	A
dequeue	_ _ C _	2	1	B
enqueue D	_ _ C D	2	2	
enqueue E	E _ C D	2	3	← wrapped to slot 0
enqueue F	E F C D	2	4	← full
dequeue	E F _ D	3	3	C

Read the block as a ring from `head`: after "enqueue F" it is C D E F — slots 2, 3, 0, 1. E and F were written into the slots that A and B vacated; no element ever moved. This trace was produced by running a reference implementation, and every row can be checked with `tail = (head + size) % 4`: before "enqueue E", tail = (2 + 2) % 4 = 0.

Full or empty?



When `head == tail`, the ring is **either** empty **or** full. Head and tail alone cannot tell which.

Look at the start of the trace (head 0, size 0, tail 0) and after "enqueue F" (head 2, size 4, tail (2 + 4) % 4 = 2). In both, the next write would go to the slot `head` points at. A ring that stores only `head` and `tail` cannot tell these two situations apart. There are three standard fixes:

1. **Keep a count** — `_size`. Empty is `size == 0`, full is `size == capacity`. This is what `dsa/queue.py` does, and why `is_empty` and `is_full` are given to you as one-liners.
2. **Leave one slot unused.** Full is defined as `(tail + 1) % capacity == head`, so `head == tail` can only mean empty. Costs one slot; common in hardware and in C, where a count shared between two threads is awkward.
3. **A flag** set on the enqueue that makes `head == tail`, cleared on the next dequeue.

Every one of them works. The bug is to have none, which is a classic exam question (W7-B3).

When the ring is full

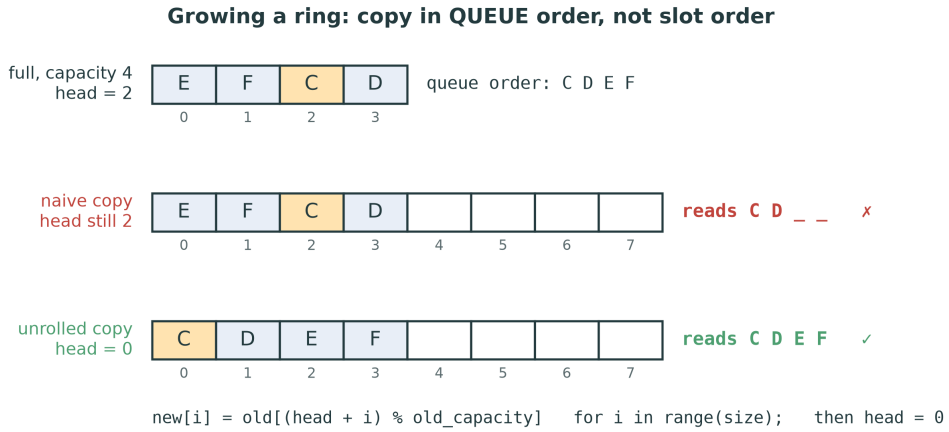
Two honest choices:

- **Refuse:** raise an error. A *bounded* queue — fixed memory, like a real buffer.
- **Grow** (the challenge): allocate a bigger `Array` and copy — as `DynamicArray` does, doubling so that enqueue stays amortised $O(1)$.

The docstring of `CircularQueue.enqueue` calls growing a **challenge**. The tests never overfill a ring, so raising `IndexError("queue is full")` is enough to pass them. But you should grow: Week 11's `level_order` and Week 14's `bfs` use your `CircularQueue`, and neither knows in advance how many items it will hold.

Both choices are used in practice. An audio driver's buffer is bounded on purpose: if the consumer falls behind, old data is dropped or the producer waits, but memory never grows. A general-purpose queue, like Python's deque, grows.

Growing: the trap



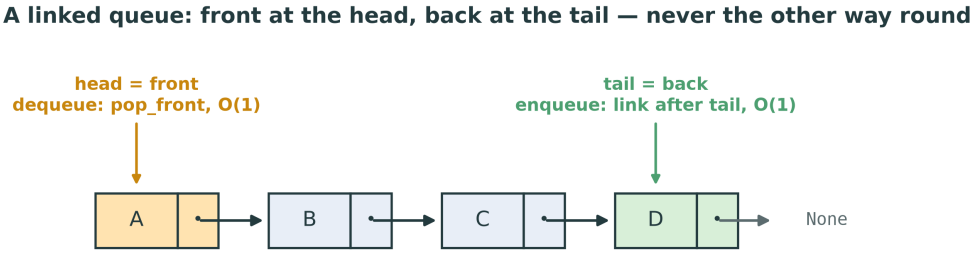
Copying slot by slot keeps the wrapped order — so the queue reads **wrong**. Copy in **queue order**, and restart head at 0.

This is the trap the docstring warns about ("a naive copy scrambles the order once the data has wrapped around"). In the figure, the full ring of capacity 4 holds C D E F starting at slot 2. Copying `new[i] = old[i]` puts E and F at slots 0 and 1 of the new block and C D at 2 and 3; with head still 2 and a capacity of 8, the queue now reads C, D, then two empty slots — E and F are lost behind the front. If the ring has not wrapped yet (head = 0), the naive copy happens to work, which is exactly why this bug survives casual testing.

The fix **unrolls** the ring while copying: the i-th item of the queue, which lives at `(head + i) % old_capacity`, goes to slot i of the new block, for i from 0 to size − 1. Then `head` becomes 0, and `capacity` the new capacity (the last is easy to forget, and then every `%` still wraps at the old size). The front goes to slot 0, and the rest follow in order. Doubling the capacity each time gives the same amortised $O(1)$ enqueue as Lecture 04: over n enqueues, the copies total less than 2n.

Two More Honest Queues

The linked queue



Front at the **head**, back at the **tail**: both operations $O(1)$ worst case.

A singly linked list with a `tail` reference can add at either end in $O(1)$, but can remove only at the head in $O(1)$: removing the last node needs the node **before** it, which takes a walk (Lecture

05, W5-K3). So the queue's front must be the head (where we remove) and its back the tail (where we add). The other way round, every dequeue is $O(n)$.

- **enqueue:** make a node. If the queue is empty, it becomes both `head` and `tail`; otherwise link it after `tail`, and it becomes the new `tail`.
- **dequeue:** `IndexError` if empty; otherwise take the head's value and move `head` to the next node.

The one trap is the end of dequeue: when the queue becomes empty, `tail` must be reset to `None` too, or it keeps pointing at a node that is no longer in the queue and the next enqueue links onto it (W7-B4).

A queue from two stacks

- `inbox`: enqueue pushes here.
- `outbox`: dequeue pops here. When `outbox` is **empty**, first pour **all** of `inbox` into it.

Pouring reverses the order — and LIFO reversed is FIFO.

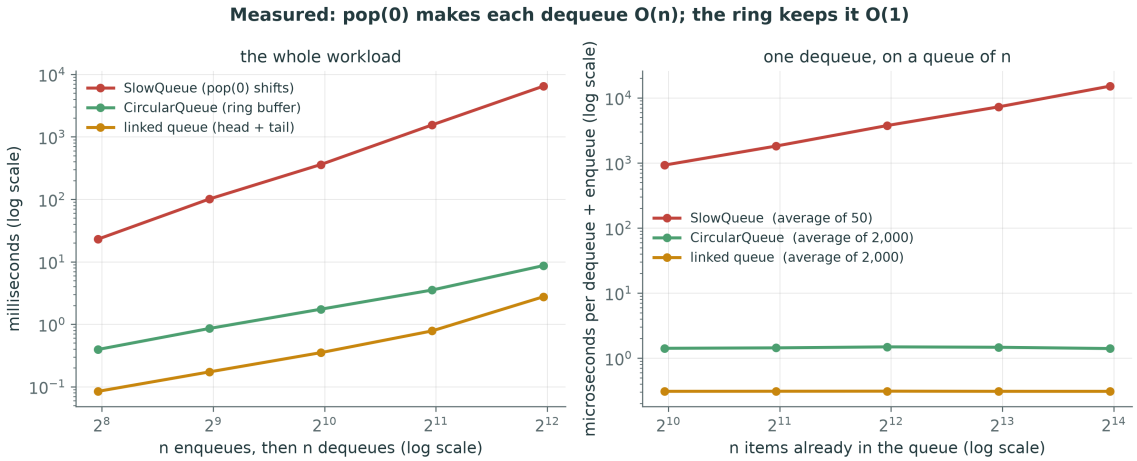
One `dequeue` may pour n items, so a single operation can cost $O(n)$. But each item is pushed onto `inbox` once, moved to `outbox` once, and popped from `outbox` once: three $O(1)$ steps over its whole life. So n operations cost $O(n)$ in total, and each is **amortised** $O(1)$ — the aggregate argument of Lecture 04 again, and of `next_greater` in W6-C3.

The rule "pour only when `outbox` is empty" is essential. Pouring on every dequeue is still correct, but each dequeue then moves everything twice, $O(n)$ each (W7-K3). It is W7-C1 in the question bank.

Why would anyone build a queue this way? In languages where the standard structures are **immutable**, as in Haskell or Erlang, a stack is a linked list that can be shared safely, and two of them make the standard functional queue. It is also a favourite interview question, and a good test of whether you understand amortised cost.

Measured

The gap is the lesson



Real timings of `dsa/queue.py` — the instructor's reference solution, built on the course `Array` exactly as yours will be — against a simple linked queue. The ring is created with room for all n items, so growing plays no part. Left: n enqueues followed by n dequeues. The two honest queues rise with slope 1 on the log–log plot, $O(n)$ for n operations; the `SlowQueue` rises with slope 2, $O(n^2)$. At n = 4,000 that was about 6.5 seconds against 9 milliseconds for the ring: some 700 times slower, and the gap doubles every time n doubles.

Right: the cost of **one** dequeue (paired with one enqueue, to keep the size at n) on a queue already holding n items. The ring and the linked queue are flat lines: $O(1)$, however long the queue. The `SlowQueue` line is a straight line of slope 1: $O(n)$ per dequeue, about 15 milliseconds at n = 16,000 against roughly 1.4 microseconds for the ring. (The $O(1)$ lines are averages over 2,000 operations, because one operation is too quick to time on its own.)

Constant factors. The linked queue beats the ring by four to five times here, because every ring operation does a `%` and goes through the Python-level bounds check of `dsa/array.py`, while a node's `next` is a plain attribute. The same effect as in Lecture 06. Big-O says both are $O(1)$; measurement says how big the 1 is.

This is the demonstration the course plan promised in Week 7: the same tests pass for both classes, and one of them is quadratic. Plot your own with the `07-queues` notebook once `dsa/queue.py` works.

Five designs, compared

	enqueue	dequeue	memory
`SlowQueue` — array, `pop(0)`	amortised $O(1)$	$O(n)$	compact
Array, head index, no wrap	amortised $O(1)$	$O(1)$	**dead slots** grow for ever
`CircularQueue` — ring buffer	$O(1)$ (amortised if it grows)	$O(1)$	compact, reuses slots
Linked, head + tail	$O(1)$	$O(1)$	a node per item

	enqueue	dequeue	memory
Two stacks	amortised $O(1)$	amortised $O(1)$	two arrays

The ring buffer is the usual choice when the maximum size is known or growth is rare: it is contiguous, it allocates nothing per item, and it never moves data except while growing. The linked queue has the best worst case, one allocation per item. Python's `collections.deque` combines the two: a doubly linked list of fixed blocks of 64 slots, so it allocates rarely, like an array, and never shifts, like a list of nodes.

Where Queues Are Used

FIFO is fairness — and breadth

- **Scheduling:** print jobs, requests to a web server, processes sharing a CPU (round-robin).
- **Buffers:** keyboard input, network packets, audio — ring buffers between a producer and a consumer; "the last k values" for a moving average.
- **Breadth-first search** (Week 14): visit everything 1 step away, then 2 steps, … — the queue keeps the frontier in order.
- **Level-order traversal** of a tree (Week 11): the same idea on a tree.

Round-robin. An operating system gives each ready process a short time slice. A process whose slice runs out goes to the back of the ready queue; the next one is dequeued from the front. Everyone gets a turn, in order. The same dequeue-then-enqueue step solves the Josephus problem — people in a circle, every k-th one leaves — which is W7-C2: put everyone in a queue, then move k − 1 people from the front to the back and remove the k-th, until one is left.

Breadth-first order. A queue processes things in the order they were *discovered*. Start from one node, enqueue its neighbours, then theirs: everything one step away is dequeued before anything two steps away. That is why BFS finds shortest paths (by number of edges), and why `level_order` in `dsa/tree.py` visits a tree row by row. Replace the queue with a stack and the same loop becomes depth-first — Week 14 puts them side by side.

A small example you can run now: the binary numbers 1 to n, in order, generated breadth-first (W7-C5). Start with "1"; each time you dequeue a string s, enqueue s + "0" and s + "1".

The last k values. A `CircularQueue(k)` that dequeues the oldest value whenever a new one arrives always holds the latest k: the window of a moving average, the last k log lines, the last k readings of a sensor. Keep a running total beside it and each new average costs $O(1)$ instead of $O(k)$ — W7-C3.

The deque: both ends, both ways

A **deque** (double-ended queue, say "deck"): add and remove at **both** ends, all $O(1)$.

- only one end used → a **stack**
- add at one end, remove at the other → a **queue**

In Python: `collections.deque` — `append`, `appendleft`, `pop`, `popleft`.

The name *deque* was coined by E. J. Schweppe, according to Knuth (*The Art of Computer Programming*, vol. 1, §2.2.1). A ring buffer makes a deque with no new ideas: adding at the front means moving `head` **back** one slot, `head = (head - 1) % capacity`. Python's `%` always returns a non-negative result for a positive capacity, so `(0 - 1) % 8` is 7: the step back from slot 0 wraps round to the last slot, exactly as needed. (In C and Java, `-1 % 8` is −1, and you must write `(head - 1 + capacity) % capacity` instead.) W7-S3 draws a deque on a ring, step by step.

Where next. Week 12 builds a **priority queue**: a queue that serves the most **urgent** item rather than the oldest one. The ADT has the same two operations, but a completely different structure underneath, the heap. Same name, different contract.

Watch it run: the Queue Lab

Queue Lab — open it in any browser.

- The code, the variables and the block drawn as a **strip and as a ring**
- `SlowQueue` shifting on every dequeue; the head index that **crawls**
- The ring buffer: this lecture's trace, a **wrap**, a ring that **grows**
- Full or empty; the linked queue; a queue from **two stacks**
- Josephus, binary numbers, a moving average, a deque on a ring
- *What happens next?* — answer before you step
- Ten scenarios that **go wrong**: no `%`, a naive grow, a stale `tail`…

The Queue Lab is the demo used in the lecture. It runs every design in this handout, step by step: the line that runs, every variable, and the storage. A `CircularQueue` is drawn twice, as the `Array` it really is and as a ring, so you can see `head` move, the tail wrap round to slot 0, and the freed slots being reused. Counters keep the slot writes and shifts, so `SlowQueue` shows its cost.

- **Online:** https://helghareeb.github.io/DSA27/lectures/07-queues/queue-lab.html
- **Offline:** open `docs/lectures/07-queues/queue-lab.html` from your copy of the repository. It is a single file and needs no installation.

How to use it:

- Pick a scenario at the top. Change the capacity, the values or the list of operations in the `>>>` line and press **Run**.
- **Step** (→) moves one step and **Back** (←) undoes it. **Skip loop** (O) runs the rest of a loop in one step; **Step out** (U) finishes the current method. **Play** (P) runs on its own.
- At the key moments the demo stops with a question. Answer it (A–D) **before** you look; that is where the learning happens.

What to try, in order: `SlowQueue`'s dequeue and the crawling head; the capacity-4 trace from the slides, then Homework 7's trace (do it on paper first); growing a wrapped ring next to the naive grow; the head-and-tail ring that cannot tell full from empty; the linked queue next to the stale `tail`; the two-stack queue next to pouring on every dequeue. For each broken scenario, find the bug in the code **before** you step.

The `SlowQueue` and `CircularQueue` scenarios show code that solves this week's exercise in `dsa/queue.py`, and the applications solve question-bank challenges. Write your own version

first. The linked queue, the crawling queue, the head-and-tail ring and the deque were written for the demo; there is no exercise for them.

This Week

Exercises: `dsa/queue.py`

Method	Target	The trap
`SlowQueue.enqueue`	amortised $O(1)$	append on `self._items`
`SlowQueue.dequeue`	$O(n)$ — on purpose	`IndexError` when empty; `pop(0)`
`CircularQueue.enqueue`	$O(1)$	tail = `(head + size) % capacity`; full?
`CircularQueue.dequeue`	$O(1)$	`IndexError` when empty; head wraps
challenge: grow when full	amortised $O(1)$	copy in queue order; head = 0

```
pytest tests/test_stack_queue.py -v -k "fifo or dequeue or circular"
```

`SlowQueue` stands on your `DynamicArray`, so `tests/test_dynamic_array.py` must pass first. `CircularQueue` stands directly on the course `Array`: its `__init__`, `is_empty`, `is_full`, `__len__`, `__iter__` and `__repr__` are given to you — read `__iter__` first, since it already contains the ring arithmetic.

Four tests cover the queues: FIFO order and the empty `IndexError` for both classes, a wrap-around at capacity 3, and ten interleaved operations that take a capacity-4 ring round twice.

Homework 7 — before Lecture 08

1. **Implement** `dsa/queue.py` until the queue tests pass — then the challenge: grow instead of raising, and test it on a ring that has wrapped.
2. **Measure** `SlowQueue` against `CircularQueue` in `notebooks/07-queues.ipynb` and plot both. Estimate how long the `SlowQueue` would take for n = 10^6.
3. **Trace** a `CircularQueue(capacity=3)` through: enqueue 1, 2, 3; dequeue; enqueue 4; dequeue; dequeue; enqueue 5, 6. Show the block, head and size after each step.
4. **Design.** A queue from two stacks: write it, and explain why it is amortised $O(1)$ (W7-C1).

For item 2, a sensible range is n = 500 to 8,000; beyond that the `SlowQueue` takes minutes. Fit the slope of your log–log line and extrapolate: slope 2 means 10 times more items cost 100 times more time.

For item 3, check your final answer with the given `__iter__`: `list(q)` should read the queue front to back.

Practise online

- **Week 7 quiz**: 40 multiple-choice questions, every answer explained

- **Written practice** on the same page: 9 questions. Write, then mark yourself against the marking points
- **Lecture 7 test**: 20 questions, 25 minutes, timed
- **Midterm simulator**: Lectures 1–7, papers of up to 60 questions
- In Arabic: week 7 · midterm

Everything runs in the browser and keeps your results on your own device. Do the quiz **after** the homework, as a check, not instead of it: it shows you which ideas did not stick, and the explanation of each answer sends you back to the right part of this handout.

- Quiz and written practice: https://helghareeb.github.io/en/quizzes/dsa27/week-07/
- Timed test on this lecture: https://helghareeb.github.io/en/quizzes/dsa27/midterm/#lecture=7

Summary

Seven things to keep

1. A queue is **FIFO**: enqueue at the **back**, dequeue at the **front**, both $O(1)$.
2. Empty dequeue raises `IndexError`.
3. On a plain array, one end is always the front: `pop(0)` or `insert(0)` — $O(n)$.
4. The **ring buffer**: head moves, nothing shifts; tail = `(head + size) % capacity`.
5. `head == tail` means **full or empty**: keep `size` (or waste one slot).
6. Growing a ring: copy in **queue order**, then `head = 0`.
7. Linked queue: front at the **head**, back at the **tail**. Two stacks: **amortised** $O(1)$.

Next

Week 8 — Searching *(midterm window)*. Seven ways to find a value, and what a **sorted** array buys you: from $O(n)$ to $O(\log n)$.

Sources and further reading

- **D. E. Knuth.** *The Art of Computer Programming*, vol. 1, *Fundamental Algorithms*, 3rd ed., Addison-Wesley, 1997, §2.2.1 "Stacks, queues, and deques" and §2.2.2 "Sequential allocation" — queues in a circular block, and the origin of the word *deque*.
- **M. T. Goodrich, R. Tamassia and M. H. Goldwasser.** *Data Structures and Algorithms in Python*, Wiley, 2013, chapter 6, "Stacks, Queues, and Deques" — an array-based circular queue that grows, written in Python.
- **T. H. Cormen, C. E. Leiserson, R. L. Rivest and C. Stein.** *Introduction to Algorithms*, 3rd ed., MIT Press, 2009, section 10.1, "Stacks and queues", and exercise 10.1-6, a queue from two stacks.
- **CPython source.** `Modules/_collectionsmodule.c` — deque as a doubly linked list of 64-slot blocks.

Every figure in this lecture is generated by `tools/figures_l07.py`. The timing figure is real measurement and will differ slightly on your machine.